

Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Sem II, 2025-2026

Tutorial-2 (Nodal and Mesh Analysis, Superposition Theorem)

The goal of this tutorial is to make a transition from relying on intuitive shortcuts to mastering systematic circuit analysis methods (e.g. Nodal/Mesh Analysis) that can help you solve electrical circuits, regardless of complexity.

Q1. You are analysing a node in a sensor network where the sensors act as Current Sources. You may choose Nodal/Mesh Analysis. Justify your choice.

- Initially, the resultant circuit is as shown in Figure.1 (a). Calculate I_o flowing in $3\text{ k}\Omega$ resistor.
- Then, the two sensors are connected as shown in Figure.1(b). Find V_1 across the $3\text{ k}\Omega$ resistor.

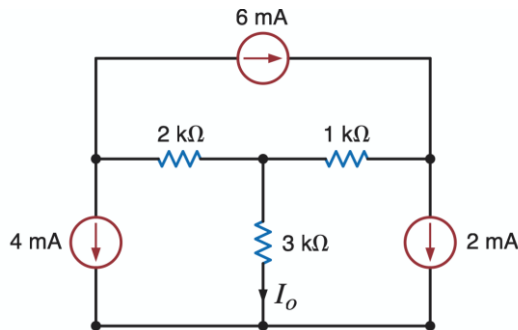


Figure 1(a)

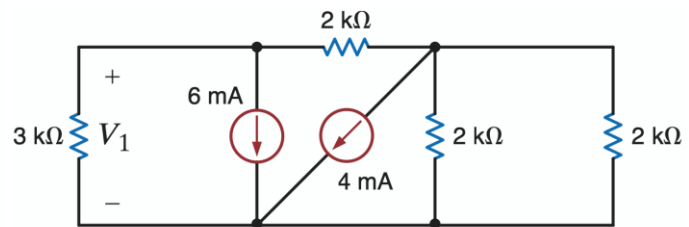
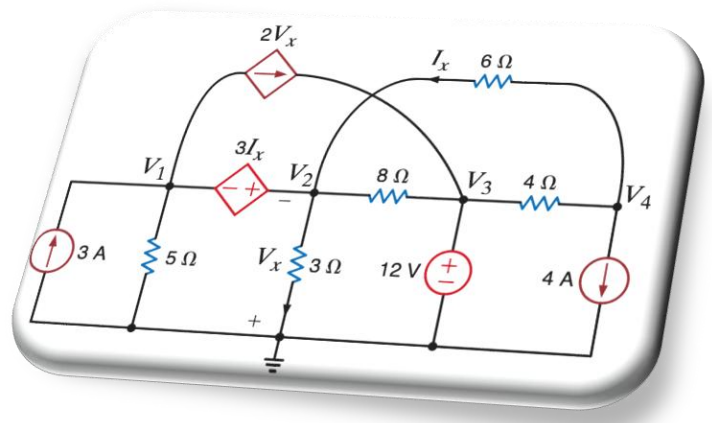


Figure 1(b)

Q2. In many amplifier and sensor circuits, a voltage source establishes a fixed potential difference between two points, neither of which is ground. This creates a "floating" source that could complicate standard nodal analysis because the current passing through the voltage source is initially unknown. The given circuit consists of current dependent voltage source.

Determine the node voltages in the given circuit.



Q3. You are analysing a representative power distribution circuit driven by independent and dependent voltage sources.

- Find the power dissipated in the $6\text{ k}\Omega$ resistor for circuit shown in A.
- Find I_1 , I_2 , I_3 and I_4 for the circuit shown in B.

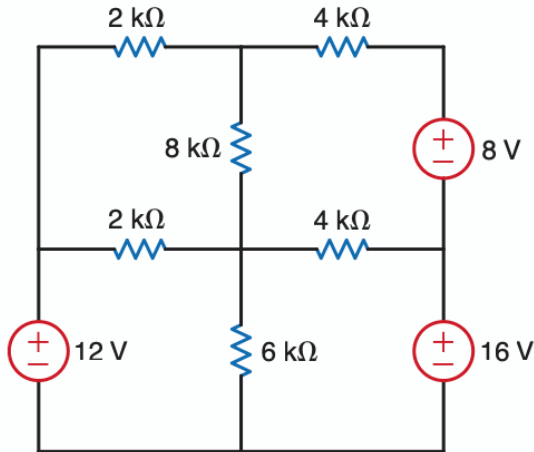


Figure A

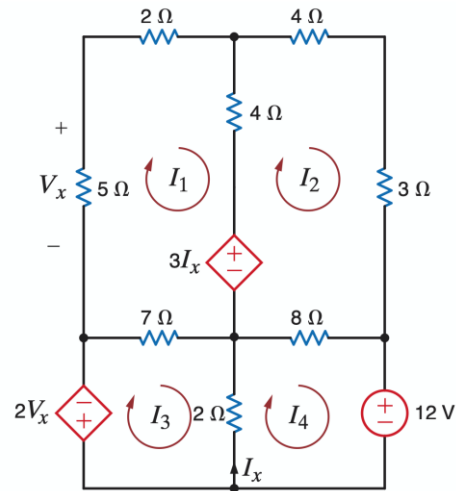
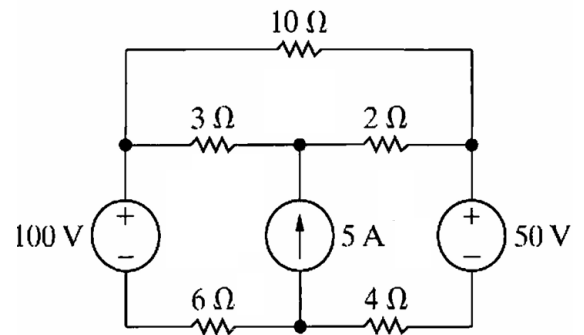


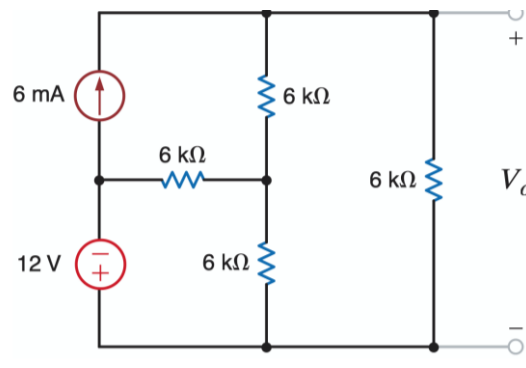
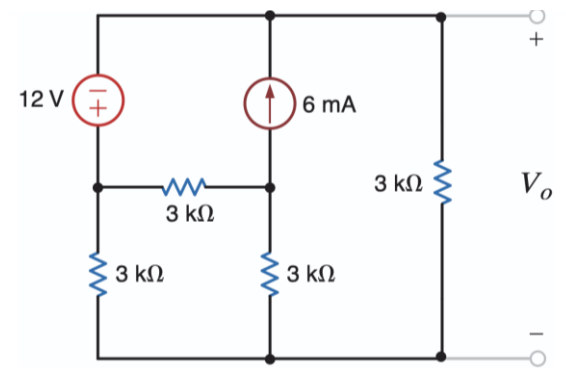
Figure B

Q4. Some precision power supplies use a current regulator shared between two control loops. When a current source sits on the boundary between two meshes, you cannot write a standard KVL loop equation because the voltage drop across an ideal current source is undefined (it adjusts to whatever the circuit needs).

For the given circuit, find the power dissipated in the $3\text{ }\Omega$ resistor.



Q5. You could encounter a circuit mixing the two different sources (e.g. a music signal modelled as a voltage source and a microphone signal modelled as a current source). Calculate the combined output voltage V_o for two given configurations using the superposition theorem.



More Practice Problems

Practice questions:

- Chapter 3 problems in Irwin J.D. and Nelms R.M., Basic Engineering Circuit Analysis.
- Chapter 4 problems in Nilsson J., S.-Riedel, Electric-Circuits, 9th-ed, Prentice-Hall-2011.